Strategic Interactions in Foreign Aid Games

POUSSE Guillaume | PEREIRA Rodrigo | SHARMA Vivan

May 6, 2024

Abstract

We present a game-theoretic examination of donor and recipient country interactions in the context of foreign aid. The study is primarily motivated by the exploitation of countries in need of international aid. To this extent, we wish to better understand what motivates a donor to provide foreign aid and when a recipient country should accept it. Donor types are restricted to strategic or altruistic throughout. We start by building a two-player, sequential baseline game, in which we demonstrate the importance of including aid-for-policy deals when modeling foreign aid interactions. We build from this result by incorporating such deals in our extended game. This first game extension is comprised of 2 sub-games: the one-period game and the two-period game, in which we show that the recipient country's gain from policy deviation must be greater than a proportion of the aid given, and also that the Recipient has a threshold probability value for which he assumes the Donor to be strategic. These conditions change as we increase the number of repetitions of the game. The strategic donor now also needs the gain from policy change to increase and the Recipient's belief that he is strategic to decrease. Concurrently, rejecting aid becomes an increasingly attractive option for the Recipient, as aid dependency keeps increasing. Lastly, To understand how competition among donors affects the previously derived equilibrium conditions, we model a first-price sealed-bid auction and calculate the optimal bids for each type of donor. We observe that a recipient country is better off than it would have been without donor competition. Through this lens, we contribute to a deeper understanding of the strategic landscape of aid provision, offering insights that inform policy formulation and enhance the efficiency and impact of international development efforts.

Department of Economics

Sciences Po - Paris

Contents

Page

1 Motivation and Literature Review **2**

2 Baseline Game **4**

2.1 Defining the Game 4

2.2 Extensive Form 5

2.3 Equilibrium 5

3 Extended Game: Policy Reform **7**

3.1 The One Period Game 7

3.1.1 Defining the Game 7

3.1.2 Extensive Form 8

3.1.3 Equilibrium 8

3.2 The Two Period Game 10

3.2.1 Defining the Game 10

3.2.2 Extensive Form 11

3.2.3 Equilibrium 11

4 Extended Game: Donor Competition **13**

4.1 Defining the Game 13

4.2 Equilibrium 13

5 Further Research **15**

5.1 Finitely Repeated Game 15

5.2 Recipient Non-Compliance Threat 16

6 Conclusion **17**

7 Appendix **18**

1 Motivation and Literature Review

The prominent rise of collaborative frameworks and diplomatic initiatives has contributed to multilateralism becoming a cornerstone of contemporary international affairs. However, global politics has seen leading nations continuously seek to extend their spheres of influence, reflecting a perennial pursuit of hegemonic power, resources, and strategic advantages. Some countries convey their intentions directly, albeit in extreme circumstances, by providing military support, declaring war, and raising or lowering bilateral trade barriers. However, when relations are ambiguous, any interaction between two countries becomes a strategic game. In this paper, we look to study one such mechanism - foreign aid.

According to Seiglie (1999), the main objectives for providing foreign aid have been economic, strategic, and humanitarian. We will briefly show how economic objectives are nested within strategic motivations, thus reducing his categorization. We then introduce humanitarian and strategic foreign aid to motivate the structure of this paper.

Kanbur (2006) studies the motives from Seiglie (1999) and shows the first, economic, to be useful in the context of tied aid and spillover effects. Interestingly, both these concepts have strategic implications. The former, tied aid, can be used to buy certain products exclusively from the donor country as a way of "redistributing income within the donor country". For example, in the second half of the 20th century, Japan's extensive aid program provided support to Asian countries employing procurement through Japanese companies, helping to fuel much of Japan's economic development (Williams, 2023). The latter, spillover effects, can produce positive or negative externalities in the context of today's globally interdependent world. Presumably, aid can foster the first kind while preventing the other (Kanbur, 2006). However, foreign aid is hardly a lucrative investment; the pure economic incentives remain limited, hence the importance of an additional strategic benefit. Today, economic aid remains one of the largest and most stable financial flows to developing countries since the second half of the 20th century.

Humanitarian aid, whereby the increased well-being of the recipient country motivates donors to provide aid, is an important argument behind international aid prevailing in the face of low economic incentives. In his influential book *Moral Vision in International Politics*, Lumsdaine (1993) combines statistical and historical data to show that international aid was driven less by donor economic and political interests but rather by humanitarian beliefs. He argues that the large contributions to international institutions over which donor countries had little influence establish foreign aid as a "response to world poverty which arose mainly from ethical and humane concerns". Thereby, altruism becomes a relevant profile type to consider in the context of international aid. Notwithstanding, we wish to point to one strong limitation to Lumsdaine (1993); its publication coincides with the end of the Cold War, a period of ideological rivalry characterized by competition for global dominance between NATO and the Soviet Union. During this period, foreign aid was used extensively as a strategic tool in the fight for global soft power.

Indeed, foreign aid can be used as a strategic tool to inculcate friendships, spread an ideology, or be used as leverage to get a foothold in a region's political landscape. The Cold War owes its emergence to three major events following the end of World War II: the implementation of the Marshall Plan, the founding of significant international organizations including the United Nations, IMF, and World Bank, and the signing of the Warsaw Pact. These events set the Cold War in motion, consequently providing the framework to study international aid in a strategic setting. Beim (1964) provides a rational basis for making foreign aid policy in the context of the Cold War at a time with little literature on interpreting the foreign aid actions of the Communist Bloc. He uses game theory to illuminate the problem of underlying self-interest in foreign aid by relying on the study of country actions rather than statements. Beim (1964) formalizes foreign aid as a game and asks: Why should a country look to give aid? and when should another country accept it?

Grant (1979) and Bueno de Mesquita and Smith (2016) contribute to the literature studying strategic foreign aid during the Cold War. Contrarily to Beim (1964), Grant (1979) does not employ formal game theory; he puts forward a qualitative argument critiquing the Cold War foreign aid regime, emphasizing the lack of political, economic, and social progress by pointing to amplified disparities between developed (donor), and developing (recipient) countries. Once more, this raises the question: when should a country

accept foreign aid? Bueno de Mesquita and Smith (2016) take a more quantitative approach to show that the United States had higher aid for policy leverage before the Soviet Union entered as a competing donor. This paper raises an interesting question on competition between donors to provide strategic aid.

We look to further motivate the study of altruistic and strategic aid in the context of more recent global events. For example, China has been garnering attention for its foreign aid and commercial construction projects across Asia and Africa, such as the Belt and Road Initiative. Ever since the takeover of Hambantota Port in Sri Lanka – caused by defaults on predatory loans taken on forecasts of profitability – alarm bells have been ringing in the ears of their rivals (Gupta, 2022). They fear this colossal infrastructure network could be converted into an unprecedented network of military bases (Frayer, 2022). Moreover, the Russo-Ukrainian and Israeli-Palestinian wars have prompted a stark division of countries on the global scale, in addition to drastically larger flows of international aid. In 2023, international assistance from official donors surged to USD 223.7 billion, marking a significant increase from the USD 211 billion reported in 2022 (OECD, 2024). This rise is partly fuelled by heightened aid contributions to Ukraine and a greater allocation of humanitarian support to developing nations by provider countries. This is no small amount of money and is especially significant when considering global struggles for soft power accumulation.

Our study aims to provide insights into the strategic interactions between donors and recipients of foreign aid. We rely on the rigorous framework provided by game theory to formally model a foreign aid game. Through this, we look to understand what drives a donor country to offer aid and a recipient country to accept it. In Section I we build a baseline game to study the conditions for a donor to offer aid and a recipient to accept. In this very simplified game, we model aid to be unconditional and show that this creates a complete information game that a strategic donor will never offer. Thus, if the donor offers, the recipient will know the aid is altruistic and will always accept it. In Section II, we look to represent the foreign aid game more accurately by assuming the donor will have the option to enforce a policy change conditional on the recipient accepting. We model the payoffs accordingly by defining utility functions to capture the relevant variables and repeat the game on a finite basis while also showing why the infinitely repeated game yields unrealistic conclusions. From this, we analyze the cases in which a recipient accepts aid and in which a donor enforces policy. Lastly, in Section III, we study the impact of donor competition by modeling the game extension as an auction between donor countries, using the payoff functions determined in Section II. In a tied competition, we assume the recipient prefers an offer from the more politically aligned donor.

2 Baseline Game

2.1 Defining the Game

The baseline game is a two player, sequential game, with incomplete information. By solving this game, we will show that the absence of aid-for-policy deals makes this a complete information game in which the donor will only ever offer if he is of humanitarian type. The purpose of this exercise is to demonstrate the importance of including aid-for-policy deals in any theoretical game model of international aid.

Definition 1. *There are two players: a donor (D) and a recipient (R).*

$$I = \{D, R\}$$

Definition 2. *There are two types of donors: strategic (θ_1) and altruistic (θ_2).*

$$\Theta = \{\theta_1, \theta_2\}$$

Definition 3. *Nature decides with probability 'q' that the recipient country is of strategic interest to the donor, and with 1-q that it is not.*

$$Pr(\Theta = \theta_1) = q \ \wedge \ Pr(\Theta = \theta_2) = 1 - q$$

Definition 4. *The Donor can either chose to offer (O) or not to offer (NO) aid, and the recipient can choose to accept (A) or reject (R).*

$$A_D = \{O, NO\} \ \wedge \ A_R = \{A, R\}$$

Order of Play

The game is played sequentially over two stages but in one time period.

In stage 1, the Donor moves first, deciding to offer (O) or not to offer(NO).

This makes the game information asymmetric, as the Donor is aware of whether the recipient is of strategic interest to it and decides accordingly.

However, the Recipient only knows the probability distribution for the same based on the timing, amount, and context of the aid offered, i.e. '*q*' and '*1-q*'.

In stage 2, the Recipient chooses whether or accept (A) or reject (R) the aid.

Assumption 1. *Both agents are rational: they seek to maximize their payoffs.*

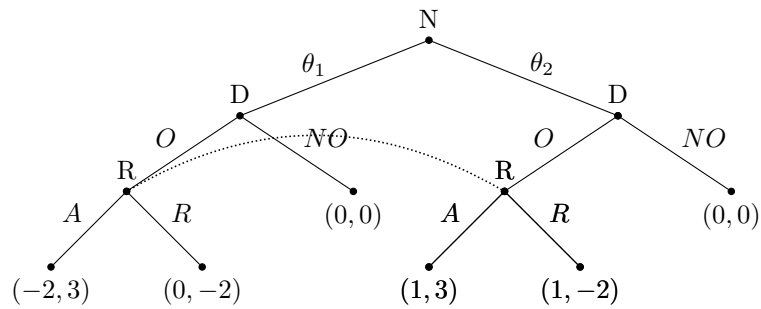
Assumption 2. *The Recipient dislikes rejecting aid when it is altruistic - a missed windfall.*

Assumption 3. *Rejecting aid from the Donor gives a positive payoff only when it is strategic – as it not only denies entry to foreign influence but also increases international repute.*

Assumption 4. *This is not a zero-sum game as both players benefit from certain scenarios, but is also not a cooperative game.*

Assumption 5. *The model considers only pecuniary payoffs, not gains or losses from policy interventions.*

2.2 Extensive Form



Here, **N** represents Nature in deciding whether the recipient, **R** is of strategic interest to the donor, **D**. The payoffs can be written as:

$$(u_D^{\theta_1}, u_R) = \begin{cases} (-2,3) & \text{at } n^{\theta_1}(A,O) \\ (0,-2) & \text{at } n^{\theta_1}(R,O) \\ (0,0) & \text{at } n^{\theta_1}(NO) \end{cases} \qquad (u_D^{\theta_2}, u_R) = \begin{cases} (1,3) & \text{at } n^{\theta_2}(A,O) \\ (1,-2) & \text{at } n^{\theta_2}(R,O) \\ (0,0) & \text{at } n^{\theta_2}(NO) \end{cases}$$

2.3 Equilibrium

We solve the game by representing it in normal form. The corresponding payoff matrix is:

Figure 1: Baseline Game Payoff Matrix

Donor

		$O_{\theta_1}, O_{\theta_2}$	$O_{\theta_1}, NO_{\theta_2}$	$NO_{\theta_1}, O_{\theta_2}$	$NO_{\theta_1}, NO_{\theta_2}$
Recipient	*Accept*	$(3, 1-3q)$	$(3q, -2q)$	$(3-3q, 1-q)$	$(0,0)$
	Reject	$(3-5q, 1-q)$	$(-2q, 0)$	$(-2q-2, 1-q)$	$(0,0)$

We know $q \in [0,1]$. Hence, we study the three cases where $q = 0, q = 1,$ or $q \in (0,1)$.

q = 0: By elimination of strictly dominated strategies, we find two Nash equilibria in $(O_{\theta_1}, O_{\theta_2}, A)$ and $(NO_{\theta_1}, O_{\theta_2}$, A). Since $\Theta = \theta_2$, we can conclude that a Donor country will always choose to offer foreign aid when he is altruistic. Also, this shows that when an altruistic donor makes an offer, the recipient will always accept.

q = 1: By elimination of strictly dominated strategies, we find two Nash equilibra in $(NO_{\theta_1}, O_{\theta_2}, A)$ and $(NO_{\theta_1}, NO_{\theta_2}$, A). Since $\Theta = \theta_1$, we can conclude that a Donor country will never choose to offer foreign aid when he is strategic. Also, this shows that if a strategic donor were to make an offer, the recipient would always accept.

q ∈ (0,1) : By elimination of strictly dominated strategies, we find one Nash equilibrium in $(NO_{\theta_1}, O_{\theta_2}, A)$. Thus, when the Recipient is unsure of the Donor's type – the state of the world is uncertain – an altruistic donor will always offer foreign aid while a strategic donor will never, and the recipient will always accept regardless of the type.

This brings us to the following Nash equilibrium solution to the game:

$$a^* = (NO_{\theta_1}, O_{\theta_2}, A)$$

We have shown that modelling a game of international aid while excluding the existence of aid-for-policy deals produces a Nash equilibrium in which a humanitarian donor always aid, a strategic donor never offers aid, and the recipient country always accepts. A recipient country will always accept unconditional aid from a donor country; he knows that a country with strategic intentions would never offer unconditional aid. Omitting aid conditionality makes this a complete information game, hence we next look to incorporate aid-for-policy deals into the baseline game.

3 Extended Game: Policy Reform

In this section, we extend our baseline game by incorporating aid-for-policy deals. In this environment, a strategic donor will provide aid only to then leverage the recipient country's dependency to demand policy reforms. Stokke (1995) accurately describes this dynamic:

"The key element is the use of pressure, by the donor, in terms of threatening to terminate aid, or actually terminating or reducing it, if conditions are not met by the recipient. Foreign aid is used as a lever to promote objectives set by the donor which the recipient would not otherwise have agreed to."

In this new environment, we look to answer the question: if a recipient country is unaware of the donor's type, when should it accept foreign aid? First, we will study the one-period version of the game. Then, following the Stokke (1995) concept of leveraging aid dependency, we will extend the game to two periods.

3.1 The One Period Game

3.1.1 Defining the Game

Definitions and assumptions 1 to 4 still hold, assumption 5 does not hold anymore. The payoffs are modeled differently. Instead of enumerating them, we formulate utility functions with the variables described below.

Definition 5. *We define τ, the amount of aid offered by the Donor country.*

Definition 6. *We define a function $Z(\tau)$ which captures the payoff from pro-social behavior. It is increasing in τ.*

Definition 7. *We define a function $\lambda(\tau)$, which captures the absolute expected opportunity cost of rejecting aid. Correctly identifying aid and rejecting it would benefit the Recipient while rejecting it when it is altruistic would hurt the Recipient. It is also increasing in τ.*

Definition 8. *We define R_D and R_R, the respective baseline utilities the Donor and Recipient countries have from their resources.*

Definition 9. *We define X_D and X_R, the respective policy objectives of the Donor and Recipient countries.*

Definition 10. *We define $(X_D - X_R)^2$, the policy change implemented by the strategic donor if the recipient were to accept his aid. This value is known by both the strategic Donor as well as the Recipient.*

Assumption 6. *There are only two policies in this model, as defined above. They are both known by the donors and the recipient.*

This assumption has strong implications, as we will show below, since it allows the recipient to, under certain conditions, identify the type of the donor based on its offer.

Assumption 7. *The amount of aid given by both strategic and altruistic donors is the same.*

$$\tau = \tau^{\theta_1} = \tau^{\theta_2}$$

Assumption 8. *The gain from rejecting strategic aid is equal to the absolute loss from rejecting altruistic aid, denoted by $\lambda(\tau)$.*

Assumption 9. *Both strategic and altruistic donors have the same baseline resource utilities.*

$$R_D = R_{D^{\theta_1}} = R_{D^{\theta_2}}$$

Assumption 10. *The altruistic Donor will never ask for reform; it obtains utility purely from its pro-social behaviour.*

Assumption 11. *For comparability, we normalize our variables below. For example, τ can be thought of as the ratio of the amount of aid to the Recipient country's GDP and X_D, X_S reflect political alignment.*

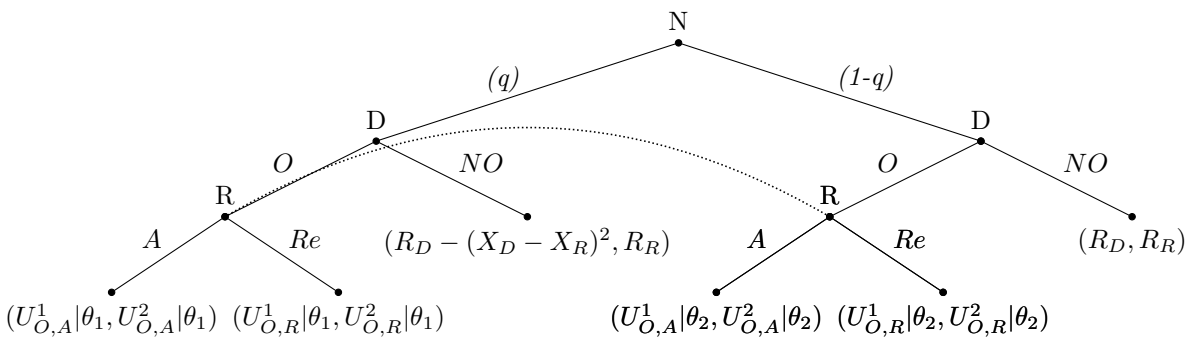
$$\tau, R_R, R_D \in [0,1] \ \wedge \ X_D, X_S \in [-1,1]$$

From the above, we build the following general utility functions[1]:

$$\begin{cases} U_O^D = R_D + I_{\theta_1}[(X_D - X_R)^2 - \tau] + I_{\theta_2} \cdot Z(\tau) \\ U_{NO}^D = R_D - I_{\theta_1} \cdot (X_D - X_R)^2 \\ U_A^R = R_R + \tau - I_{\theta_1} \cdot (X_D - X_R)^2 \\ U_R^R = R_R + I_{\theta_1} \cdot \lambda(\tau) - I_{\theta_2} \cdot \lambda(\tau) \end{cases}$$

Here, I_{θ_1} is an indicator function that takes the value 1 if θ_1 is true, and I_{θ_2} takes the value 1 if θ_2 is true.

3.1.2 Extensive Form



Where the expanded form of the above utilities are:

$$\begin{cases} (U_{O,A}^D|\theta_1, U_{O,A}^R|\theta_1) = (R_D + (X_D - X_R)^2 - \tau, R_R - (X_D - X_R)^2 + \tau) \\ (U_{O,R}^D|\theta_1, U_{O,R}^R|\theta_1) = (R_D - (X_D - X_R)^2, R_R + \lambda(\tau)) \\ (U_{O,A}^D|\theta_2, U_{O,A}^R|\theta_2) = (R_D + Z(\tau), R_R + \tau) \\ (U_{O,R}^D|\theta_2, U_{O,R}^R|\theta_2) = (R_D + Z(\tau), R_R - \lambda(\tau)) \end{cases}$$

3.1.3 Equilibrium

To solve for equilibrium, we first represent the game in normal form like we did for the baseline. The payoff matrix can be found in Appendix A. From this, we calculate two subgame-perfect Nash equilibria;

$$(Eq.1) = \begin{cases} O_{\theta_1} O_{\theta_2} & \text{if } (X_D - X_R)^2 > \dfrac{\tau}{2} \\ Accept & \text{if } q^* < \dfrac{\tau + \lambda(\tau)}{2\lambda(\tau) + (X_D - X_R)^2} \end{cases} \qquad (Eq.2) = \begin{cases} NO_{\theta_1} O_{\theta_2} & \text{if } (X_D - X_R)^2 < \dfrac{\tau}{2} \\ Accept & \text{if } \tau > \lambda(\tau) \end{cases}$$

[1]The subscripts denote the action while the superscript denote the player.

Intuitively, we see that if the utility differential from changing policies is greater than half the utility from the amount of aid given, the Recipient accepts. This, if his belief that the Donor is strategic is less than the ratio of utility received from the combined value of aid and the ideological value of rejecting it, and the sum of the latter and the policy reform. Furthermore, if the utility differential is less than half the amount of aid given, the recipient will accept it if the amount of aid given is more than the absolute ideological value of rejecting it.

It is fascinating to see that if a country is not strapped for investment, they are likely to set this ideological value to zero, thereby increasing the expected value of rejecting aid. The threshold value for q^* will increase as it decreases in $\lambda(\tau)$ and will equal the ratio of aid and policy difference. As a country grows wealthier, they are less likely to accept aid!

In equilibrium 2, the recipient will then always accept aid, as it will know that the humanitarian donor does not care about policy differences and will set $X_D = X_R$. This follows from our baseline model.

We finalize this analysis by investigating whether the recipient can infer anything about the type of donor from the offers it receives. We do this by deriving a condition for the optimal amount of aid offered by the strategic donor.

In order to calculate the optimal amount of aid the strategic donor will offer, we begin by showing that, in equilibrium, the recipient must be indifferent between accepting or rejecting the offer (here, we assume the recipient knows that the donor is strategic). We can imagine a scenario where the donor offers one euro more than the amount that makes the recipient indifferent. Obviously, the recipient would accept this offer as well. However, the donor can improve its utility by offering only half a dollar more, instead of one dollar. We can repeat this logic and eventually get that, by equalling the recipient's payoffs from accepting and rejecting, the following condition must hold in equilibrium:

$$\tau - \lambda(\tau) = (X_D - X_R)^2$$

This condition states that, in equilibrium, the amount of aid τ must equal the cost of accepting the aid, in terms of policy adoption and cost of accepting aid from a strategic donor.

We can now set up the Lagrangian to calculate the optimal bid for the strategic donor. We maximize the strategic donor's utility from offering (and seeing its offer accepted), subject to the condition specified in Eq.1 and the binding constraint we have just derived:

$$\mathcal{L} = [R_D + (X_D - X_R)^2 - \tau)] - \mu[\tau - \lambda(\tau) - (X_D - X_R)^2] - \gamma \cdot [\frac{\tau}{2} - (X_D - X_R)^2]$$

Taking the derivative with respect to τ and setting it equal to zero to maximize the expected utility, we get that:

$$\lambda'(\tau) = \frac{1 + \mu + \frac{\gamma}{2}}{\mu}$$

Since $\lambda(\tau)$ is not known by the donor, it cannot solve this problem. Therefore, it cannot rationalize an optimal offer because it lacks some crucial information about the recipient. In turn, the recipient cannot correctly predict the offer a strategic donor would make. We end by concluding that, since the offer is somewhat arbitrary within a given interval, the recipient cannot infer anything about the type of the donor strictly from the offers it receives.

3.2 The Two Period Game

Still following from Stokke (1995) on leveraging aid dependency, we extend the above aid-for-policy extension of the baseline game, to two periods. Studying the game across two periods is relevant in the context of aid distributed across time rather than a lump-sum transfer. What are the effects of providing foreign aid across time, rather than as a one time transfer?

3.2.1 Defining the Game

All assumptions from the one period game still hold, we add to them below:

Definition 11. *We define a new action for the strategic Donor in the second period: to enforce or waive the policy reform. Choosing to waive is the same as making an offer with no conditions attached. Hence:*

$$A^1_{\theta_{1,2}} = \{O, NO\} \wedge A^2_{\theta_1} = \{E, W\}$$

Definition 12. *We define a constant $\alpha \in [0.1]$, the recipient country's coefficient of aid dependency.*

Assumption 12. *Aid dependency grows over time.*

Assumption 13. *Instead of always enforcing the reform in period one, the strategic Donor has the option to waive the reform in the first period and enforcing in the second period. Strategic donors have incentive to do so.*

Assumption 12 follows from Assumption 11; as aid dependency increases over time, the Donor accumulates greater leverage over the policy-making institutions of the Recipient.

Assumption 14 *The amount of aid offered, τ, and recipient beliefs, q, are subject to change between periods 1 and 2. Hence we apply time subscripts to distinguish between τ_1, τ_2, q_1, and q_2.*

Assumption 15 *The altruistic Donor will always make an offer, regardless of the period. This follows from our previous results. We use this to eliminate redundant strategy sets for the Donor.*

Assumption 16 *The probability q is independent across time periods. Hence, a donor can switch types from one period to the next.*

$$q_1 \perp\!\!\!\perp q_2$$

Assumption 17 *While τ_1 and τ_2 may differ, both kinds of donors give the same amount of aid in each period.*

$$\tau_1^{\theta_1} = \tau_1^{\theta_2} \wedge \tau_2^{\theta_1} = \tau_2^{\theta_2}$$

Assumption 18 *The extent of policy reform is equally split over both periods; a Strategic donor gains a half payoff if he enforces policy in the first period and a complete payoff if the Recipient accepts in the second.*

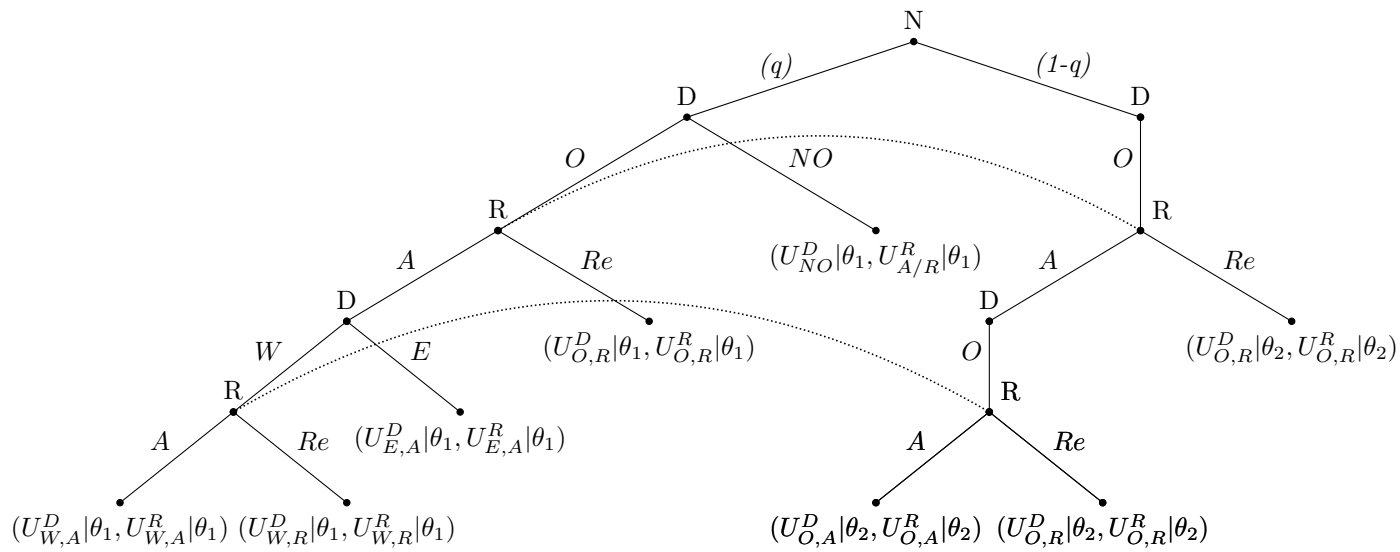
Finally, there are two game-ending actions:

1. If the Donor enforces the policy, the game ends as it reveals their type.
2. Similarly, if the Recipient rejects aid, the game ends.

The general utility functions are represented below, for Y periods that the Recipient receives aid out of N periods. In this game, we keep $Y \in \{1, 2\}$ and $N = 2^2$:

$$\begin{cases} U_A^R = R_R + \sum_{t=1}^{Y} \tau_t - I_{\theta_1} \cdot [\sum_{t=1}^{Y} \alpha\tau_{t-1} + \frac{Y}{N}(X_D - X_R)^2] \\ U_R^R = R_R + \sum_{t=1}^{Y-1} \tau_t - +I_{\theta_1} \cdot \lambda(\tau_t) - I_{\theta_2} \cdot \lambda(\tau_t) \\ U_E^D = R_D - \sum_{t=1}^{Y} \tau_t + \frac{Y}{N}(X_D - X_R)^2 \\ U_W^D = R_D - I_{\theta_1} \cdot [\sum_{t=1}^{Y} \tau_t + \frac{Y}{N}(X_D - X_R)^2] + I_{\theta_2} \cdot \sum_{t=1}^{Y} Z(\tau_t) \end{cases}$$

3.2.2 Extensive Form



The payoffs at each node are shown by:

$$\begin{cases} (U_{NO}^D|\theta_1, U_{A/R}^R|\theta_1) = (R_D - (X_D - X_R)^2, R_R) \\ (U_{O,R}^D|\theta_1, U_{O,R}^R|\theta_1) = (R_D - (X_D - X_R)^2, R_D + \lambda(\tau_1)) \\ (U_{O,R}^D|\theta_2, U_{O,R}^R|\theta_2) = (R_D + Z(\tau_1), R_R - \lambda(\tau_1)) \\ (U_{E,A}^D|\theta_1, U_{E,A}^R|\theta_1) = (R_D - \tau_1 + \frac{1}{2}(X_D - X_R)^2, R_R + \tau_1 - \frac{1}{2}(X_D - X_R)^2) \\ (U_{W,A}^D|\theta_1, U_{W,A}^R|\theta_1) = (R_D + (X_D - X_R)^2 - (\tau_1 + \tau_2), R_R + (\tau_1 + \tau_2 - \alpha\tau_1) - (X_D - X_R)^2) \\ (U_{O,A}^D|\theta_2, U_{O,A}^R|\theta_2) = (R_D + Z(\tau_1) + Z(\tau_2), R_R + \tau_1 + \tau_2) \\ (U_{O,R}^D|\theta_2, U_{O,R}^R|\theta_2) = (R_D + Z(\tau_1) + Z(\tau_2), R_R + \tau_1 - \lambda(\tau_2)) \end{cases}$$

3.2.3 Equilibrium

We solve this game exactly how we solved the last - by creating a normal form game and comparing payoffs. Key equations for this can be found in Appendix B.

[2]The subscripts denote the action while the superscript denote the player.

For the Donor, we find that the strategy of $NO_{\theta_1}O_{\theta_2}$ (Not offering when strategic, but always offering when altruistic) is weakly dominated, under the following condition:

$$(X_D - X_R)^2 > 2\tau_1 \tag{1}$$

Next, we see that the Donor will prefer $OW_{\theta_1}O_{\theta_2}$ over $OE_{\theta_1}O_{\theta_2}$ (offering and waiving over enforcing) if two conditions are met:

$$\begin{cases} q_2^* < q_1 \\ q_2^* > \dfrac{q_1[\frac{1}{2}(X_D - X_R)^2 - (\tau_1 - R_D)] + R_D}{(X_D - X_R)^2 - (\tau_1 + \tau_2)} \end{cases}$$

Given that q_2^* is a probability, it must be greater than zero. Since we also must have that $R_D \geq \tau_1$ (value of initial endowment must be more than aid offered), the numerator of (4) will always be positive. Hence, to make the entire term positive, the denominator must also be positive, implying:

$$(X_D - X_R)^2 > (\tau_1 + \tau_2) \tag{2}$$

These are significant results! They imply that for the strategic Donor to want to mislead the Recipient in period one instead of enforcing the policy, it must be that the Recipient's belief that the Donor is strategic be **decreasing over time**. Further, the gain from the change in policies for the Donor must be greater than the total aid given over the two periods, which is also an intuitive result. The strategic Donor plays only to get rewarded by a change in policies, so his expected gain must be larger than his investment for him to want to invest.

Comparing (5) with (2), we now get:

$$\tau_2 \geq \tau_1 \tag{3}$$

An implicit condition for $OW_{\theta_1}O_{\theta_2}$ to be the dominant strategy for the Donor is that the aid given in the second period must be at least that of the first period.

Coming to the Recipient, we find that the Recipient always accepts when two conditions are met:

$$q_2^* < \frac{\tau_2 + \lambda(\tau_2)}{\alpha\tau_1 + (X_D - X_R)^2 + 2\lambda(\tau_2)} \tag{4}$$

$$q_2^* > \frac{\lambda(\tau_1)(2q_1 - 1) - \tau_1 + \lambda(\tau_2)}{2\lambda(\tau_2)} \tag{5}$$

Given that q_2^* is a probability, it must be less than one and greater than zero. Using these inequalities in the two conditions respectively gives us:

$$\tau_2 - \alpha\tau_1 \leq (X_D - X_R)^2 + \lambda(\tau_2) \tag{6}$$

$$q_1^* \geq \frac{\tau_1 - \lambda(\tau_2)}{2\lambda(\tau_1)} + \frac{1}{2} \tag{7}$$

Applying the same logic for q_1^* that it must be less than one, we get:

$$\tau_1 \leq \lambda(\tau_1) + \lambda(\tau_2) \tag{8}$$

Equation (9) tells us that the loss from aid dependency must be less than the loss from policy change for the Recipient always to accept. As time goes on, aid dependency is likely to increase, which means that rejecting aid also becomes an increasingly attractive option. If all these conditions hold, we get a Nash equilibrium in $OW_{\theta_1}O_{\theta_2}, AA$.

4 Extended Game: Donor Competition

In this section, we model a different extension of the game by creating competition between two donors. To do so, we build a first-price sealed-bid auction such that two donors make one bid to offer aid to the recipient country. We derive the optimal strategy of both donors and show that, by introducing competition, the recipient can, in most situations, infer the type of donor based on the bids it receives.

4.1 Defining the Game

There are three players: two donors with different types (θ_1 with probability q and θ_2 with probability 1-q, as before) and one recipient. The utilities are the same as before, but we specify the utility of the altruistic donor further to solve for the optimum bid.

Donors compete to give aid to the Recipient. They know their own types but the recipient only has beliefs regarding the donor type, represented by a uniform probability distribution. As before, a strategic donor always offers an aid-for-policy deal. In other words, he offers a financial transfer of amount b_{θ_1} conditional on the implementation of a certain policy X_D. An altruistic donor always offers an unconditional financial transfer b_{θ_2}.

Both bids are made simultaneously. The recipient then has to decide which bid to accept (if any). The game is played only once.

The general utility functions are represented as the following:

$$\begin{cases} U_O^D = R_D + I_{\theta_1} \cdot [(X_D - X_R)^2 - b_{\theta_1}] + I_{\theta_2} \cdot [-(b_{\theta_2} - k)^2 + c] \\ U_A^R = R_R + \tau - I_{\theta_1} \cdot (X_D - X_R)^2 \\ U_R^R = R_R + I_{\theta_1} \cdot \lambda(b_\theta) - I_{\theta_2} \cdot \lambda(b_\theta) \end{cases}$$

Where:

Definition 13. *b_{θ_1} and b_{θ_2} are the bids made by the strategic and altruistic types;*

Definition 14. *k is the bid that maximizes the altruistic donor's utility. We model the utility of the altruistic donor to have a ceiling on the bid it is willing to do;*

Definition 15. *c is a constant we use to ensure the altruistic donor has non-zero utility;*

It is straightforward to check that the maximum bid for each donor depends on its type. In particular, the maximum bid b_{θ_i} is:

$$\begin{cases} v_{\theta_1} = [(X_D - X_R)^2] & \text{if } i = 1 \\ v_{\theta_2} = k & \text{if } i = 2 \end{cases}$$

4.2 Equilibrium

To find the Nash equilibrium of this auction, we start by assuming that the strategic bidder bids half his valuation: $b_{\theta_1} = 1/2 * (X_D - X_R)^2$. On the other hand, we assume the altruistic donor bids $b_{\theta_2} = s_1$. Hence, we have:

$$\begin{cases} v_{\theta_2} - b_{\theta_2} & \text{if } b_{\theta_2} > b_{\theta_1} \\ 0 & \text{if } b_{\theta_2} < b_{\theta_1} \end{cases}$$

We can calculate the expected value of the bid:

$$\mathbb{E}[b_{\theta_2}] = (v_{\theta_2} - b_{\theta_2}) \cdot \Pr(b_{\theta_2} > b_{\theta_1}) = (v_{\theta_2} - b_{\theta_2}) \cdot \int_0^{b_{\theta_1}} 1 \, dv_{\theta_1} = (v_{\theta_2} - b_{\theta_2}) \cdot 2 \cdot (b_{\theta_2}) = 2 \cdot (v_{\theta_2}) \cdot (b_{\theta_2}) - 2 \cdot (b_{\theta_2})^2$$

In order to calculate the bid that maximizes the $\mathbb{E}[b_{\theta_2}]$, we derive it with respect to b_{θ_2} and equal it to 0:

$$\frac{\partial \mathbb{E}[b_{\theta_2}]}{\partial b_{\theta_2}} = 0$$

Setting the derivative to zero will give us the bid that maximizes the expected value since the function is continuous and:

$$\frac{\partial^2 \mathbb{E}[b_{\theta_2}]}{\partial b_{\theta_2}^2} < 0$$

Therefore, we get that:

$$b^*_{\theta_2} = \frac{1}{2} \cdot k$$

Since the problem is symmetrical, we can infer that the optimum bid for the strategic player is:

$$b^*_{\theta_1} = \frac{1}{2} \cdot [(X_D - X_R)^2]$$

According to assumption 6, the recipient knows X_D and X_R. Therefore, the recipient can identify the type of donor based on the bid, as long as $k \neq (X_D - X_R)^2$. This contrasts with the policy-reform game, where the recipient was unable to identify the type of donor. Therefore, by adding competition to the game, the recipient is better off since it does not take the risk of accepting aid from a strategic donor nor rejecting aid from an altruistic donor.

As in the baseline model, the recipient will reject the bid from the strategic donor and accept the bid from the altruistic donor:

- $U_A^R|\theta_1 = R_R - \frac{1}{2} \cdot [(X_D - X_R)^2] < U_R^R|\theta_1 = R_R + \lambda(b_\theta)$
- $U_A^R|\theta_2 = R_R + \frac{1}{2} \cdot k > U_R^R|\theta_2 = R_R - \lambda(b_\theta)$

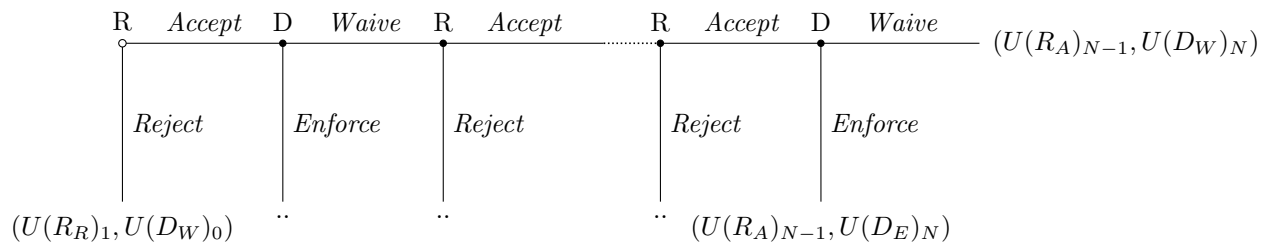
5 Further Research

In this section, we propose ideas for further research into the topic using our model.

5.1 Finitely Repeated Game

An interesting scope for further research would be to repeat the extended game with utility functions for N periods, wherein the Donor assumes complete control of the Recipient's policy-making institutions. We motivate this by using historical aid transfers as inspiration. Foreign aid is rarely a one-time lump sum transfer but is distributed over several periods. From the two-period game, we see that the equilibrium conditions for accepting and offering aid change drastically from the one-period game. Using the same logic, the conditions should change as we keep increasing stages of repetition.

We propose a simple model using the same general utility functions as in the two-period game, in the framework of a centipede game.



Where the expanded forms of the utilities at the final node are:

$$\begin{cases} (U(R_A)_{N-1}, U(D_E)_N) = (V(R_R) + \sum_{t=1}^{N} \tau_t - \sum_{t=1}^{N} \alpha\tau_{t-1} + (X_D - X_R)^2, V(R_D) - \sum_{t=1}^{N} \tau_t + (X_D - X_R)^2) \\ (U(R_A)_{N-1}, U(D_W)_N) = (V(R_R) + \sum_{t=1}^{N} \tau_t, V(R_D) + \sum_{t=1}^{Y} Z(\tau_t)) \end{cases}$$

We solve this repeated game using backward induction. However, it is important to note that for the Donor, always passing is only an equilibrium strategy if they are altruistic. In case they are strategic, enforcing in the first period is not a perfect Bayesian equilibrium (PBE) as they always stand to gain from enforcing later on. Therefore, they try to deviate at a later stage by deceiving the Recipient.

In period N-1, the Recipient will be indifferent between accepting and rejecting aid at the following value of its belief q^*:

$$q^*_{N-1} = \frac{\lambda(\tau_{N-1}) \cdot (I_{\theta_1} - I_{\theta_2}) - \tau_N}{(X_D - X_R)^2 - \sum_{t=1}^{N} \alpha\tau_{t-1}} \tag{9}$$

Where τ_N represents the amount of aid in period N.

For this result to hold, we need that q^* follows the rules of a probability distribution, $\in [0, 1]$. This means that it must be greater than zero:

$$\text{Case 1.} = \begin{cases} \lambda(\tau_{N-1}) \cdot (I_{\theta_1} - I_{\theta_2}) \leq \tau_N \\ (X_D - X_R)^2 < \sum_{t=1}^{N} \alpha\tau_{t-1} \end{cases} \qquad \text{Case 2.} = \begin{cases} \lambda(\tau_{N-1}) \cdot (I_{\theta_1} - I_{\theta_2}) \geq \tau_N \\ (X_D - X_R)^2 > \sum_{t=1}^{N} \alpha\tau_{t-1} \end{cases}$$

It also must hold for both cases that $q^*_{N-1} < 1$:

$$\lambda(\tau_{N-1}) \cdot (I_{\theta_1} - I_{\theta_2}) - \tau_N \leq (X_D - X_R)^2 - \sum_{t=1}^{N} \alpha \tau_{t-1}$$

As argued previously, $\lambda(\tau)$ is going to keep decreasing as the Recipient grows wealthier with time. Therefore, it is most likely for Case 1 to hold. Moreover, we can use this value of q^*_{N-1} to successively solve for threshold values for probabilities of the Donor to waive when he is strategic to convince the Recipient that he is altruistic. Going back to the first period would then tell us the perfect Bayesian equilibrium strategies for both players.

5.2 Recipient Non-Compliance Threat

Another potential tangent worth exploring is seeing how dynamics change if the Recipient introduces a threat to not comply with the policy change. Hence, the strategic Donor would add a probability distribution to their strategies to account for this uncertainty. This may alter their equilibrium strategies in all of the games discussed above which involve policy reform.

We think of this because of real-life instances where countries have suddenly begun rejecting foreign aid and influence, usually after a regime change. Albeit this may not be very likely to happen in a game with only two players, it may be worth considering when two strategic donors are competing for influence.

6 Conclusion

This paper offers a game-theoretic formalization of strategic interactions in foreign aid. We find that while an altruistic donor has an incentive to always offer aid, certain conditions need to be met for the strategic donor to do so. In the baseline game, we see that in the absence of aid-for-policy deals, an altruistic donor will never make an offer. Therefore, if an offer is made, the Recipient knows the Donor is altruistic and will always accept.

Following this result, we extend the baseline game by including aid-for-policy deals. We also build utility functions to model more realistic payoffs. In this game, we find that for the strategic donor to make an offer, the recipient country's gain from policy deviation must be greater than a proportion of the aid given, $\frac{\tau}{2}$, to be exact. We also find that the Recipient has a threshold probability value for which he assumes the Donor to be strategic. This value is a function of the aid being offered, the value of rejecting it, as well as the value of the change in policies. As countries grow wealthier and become increasingly indifferent to rejecting aid, this threshold becomes the ratio of aid to the absolute value of the policy change. We also attempt to estimate the strategic donor's optimal offer of aid and observe that they cannot calculate it due to lack information on the Recipient. In turn, the Recipient cannot infer the Donor's type from the offer that has been made.

We build on this by exploring the effect of repeating the game. We add a new set of actions for the strategic donor; to enforce the policy reform or to waive it to a later date – enforcing it in the last period. We also build off the results of our previous games and remove the redundant option where an altruistic donor does not offer aid. In brief, the results show that the conditions for a strategic donor to offer or waive aid and a recipient to accept or reject aid, change. Over time, the strategic donor now needs the gain from policy change to increase and the Recipient's belief on his type to decrease. The amount of aid must also increase in each period. The Recipient's conditions for accepting aid in each period are modified as well. An implicit condition is that rejecting aid becomes an increasingly attractive option over time, as aid dependency keeps increasing.

In our final extension, we seek to understand how competition among donors affects the previously derived equilibrium conditions. We model this as a first-price sealed-bid auction and calculate the optimal bids for each type of donor. Since the Recipient knows these values, it can identify the two types of donors, as long as the bids are different. Therefore, it rejects the bid from the strategic donor and accepts that of the altruistic donor, making it better off than it would have been without donor competition.

Lastly, there are some potential concerns with our study. We make many simplifying assumptions, of which some could be argued against. For example, we assume the Recipient knows the strategic donor's policy demand. However, this is not likely as different donors will have different priorities and are likely to conceal their policy intentions. We also assign quantitative values to qualitative variables for the sake of comparability, which could be considered as an oversimplification. However, the main results are useful; they successfully map real-world interactions in the foreign aid game and are likely to hold if taken through further research.

7 Appendix

Appendix A - Solving for equilibrium in one-period policy reform extension

The payoff matrix:

Table 1: Altered Payoffs

R(P1)/D(P2)	$O_{\theta_1}O_{\theta_2}$	$O_{\theta_1}NO_{\theta_2}$	$NO_{\theta_1}O_{\theta_2}$	$NO_{\theta_1}NO_{\theta_2}$
Acc	$(V(R_R) - q(X_D - X_R)^2 + \tau, V(R_D) + q(X_D - X_R)^2 - q\tau + (1-q)Z(\tau))$	$(\mathrm{V(R}_R) - (X_D - X_R)^2 + \tau, V(R_D) + q(X_D - X_R)^2 - q\tau)$	$(\mathrm{V(R}_R) + \tau(1-q), V(R_D) - q(X_D - X_R)^2 + (1-q)Z(\tau))$	$(\mathrm{V(R}_R), V(R_D) - q(X_D - X_R)^2)$
Rej	$(\mathrm{V(R}_R) - (1-2q)\lambda(\tau), V(R_D) - q(X_D - X_R)^2 + (1-q)Z(\tau))$	$(\mathrm{V(R}_R) + q\lambda(\tau), V(R_D) - q(X_D - X_R)^2)$	$(\mathrm{V(R}_R) + (1-q)\lambda(\tau), V(R_D) - q(X_D - X_R)^2 + (1-q)Z(\tau))$	$(\mathrm{V(R}_R), V(R_D) - q(X_D - X_R)^2)$

Upon comparing payoffs, we find two dominant strategies for the Donor and check the conditions for the Recipient to accept both.

Appendix B - Solving for equilibrium in two-period policy reform extension

After computing the payoff matrix, we get the following equations for getting equations (2)-(4),(7) and (8):

For (2), we compare the Donor's payoffs between $OE_{\theta_1 O_{\theta_2}}$ and $NO_{\theta_1} O_{\theta_2}$. For $NO_{\theta_1} O_{\theta_2}$ to be dominated, this inequality needs to hold:

$$V(R_D)(1+q_1-q_2)+q_1(\frac{(X_D-X_R)^2}{2}-\tau_1)+(1-q_2)(Z(\tau_1)+Z(\tau_2)) > V(R_D)(1+q_1-q_2)-q_1(X_D-X_R)^2+(1-q_2)(Z(\tau_1)+Z(\tau_2))$$

$$\implies \frac{(X_D-X_R)^2}{2} > \tau_1$$

For (3), we compare the Donor's payoffs between $OW_{\theta_1 O_{\theta_2}}$ and $OE_{\theta_1 O_{\theta_2}}$ under the Recipient's strategy of AR. For waiving to be a dominant strategy, this inequality needs to hold:

$$V(R_D) - q_2(\tau_1 - \frac{(X_D-X_R)^2}{2}) > V(R_D)(1+q_1-q_2) + q_1\frac{(X_D-X_R)^2}{2} - q_1\tau_1$$

$$\implies q_2(\tau_1 - \frac{(X_D-X_R)^2}{2} - V(R_D)) < q_1(\tau_1 - \frac{(X_D-X_R)^2}{2} - V(R_D))$$
$$\implies q_2^* < q_1^*$$

For (4), we compare the Donor's payoffs between $OW_{\theta_1 O_{\theta_2}}$ and $OE_{\theta_1 O_{\theta_2}}$ under the Recipient's strategy of AA. For waiving to be a dominant strategy, this inequality needs to hold:

$$q_2[(X_D-X_R)^2 - \tau_1 - \tau_2] > V(R_D)(1+q_1-q_2) + q_1(\frac{(X_D-X_R)^2}{2} - \tau_1$$

$$\implies q_2^* > \frac{q_1[\frac{(X_D-X_R)^2}{2} - (\tau_1 - V(R_D))] + V(R_D)}{(X_D-X_R)^2 - (\tau_1+\tau_2)}$$

For (7), we compare the Recipient's payoffs between AA and AR under the Donor's strategy of $OW_{\theta_1 O_{\theta_2}}$. For accepting to be a dominant strategy, this inequality needs to hold:

$$V(R_R) + (\tau_1+\tau_2) - q_2(\alpha\tau_1 + (X_D-X_R)^2) > V(R_R) + \tau_1 + \lambda(\tau_2)(2q_2-1)$$

$$\implies \tau_2 + \lambda(\tau_2) > q_2(\alpha\tau_1 + (X_D-X_R)^2 + 2\lambda(\tau_2))$$
$$\implies q_2^* < \frac{\tau_2+\lambda(\tau_2)}{\alpha\tau_1 + (X_D-X_R)^2 + 2\lambda(\tau_2)}$$

For (8), we compare the Recipient's payoffs between AA and R under the Donor's strategy of $OW_{\theta_1 O_{\theta_2}}$. For accepting to be a dominant strategy, this inequality needs to hold:

$$V(R_R) + \tau_1 + \lambda(\tau_2)(2q_2-1) > V(R_R) + \lambda(\tau_1)(2q_1-1)$$

$$\implies 2q_2 - 1 > \frac{\lambda(\tau_1)(2q_1-1)-\tau_1}{\lambda(\tau_2)}$$

References

David Beim. The communist bloc and the foreign aid game. *Western Political Quarterly*, 17(4):784–799, 1964.

Bruce Bueno de Mesquita and Alastair Smith. Competition and collaboration in aid-for-policy deals. *International Studies Quarterly*, 60(3):413–426, 2016.

Lauren Frayer. Why a Chinese Ship's Arrival in Sri Lanka has Caused Alarm in India and the West. https://www.npr.org/2022/08/19/1118113095/sri-lanka-china-ship-hambantota-port#:~:text=The%20Hambantota%20port%2C%20with%20its,on%20a%2099%2Dyear%20lease, 2022. [Accessed 03-05-2024].

James P Grant. Perspectives on development aid: World war ii to today and beyond. *The ANNALS of the American Academy of Political and Social Science*, 442(1):1–12, 1979.

Shishir Gupta. Was cash strapped Sri Lanka duped by China in Hambantota Port? https://www.hindustantimes.com/world-news/was-cash-strapped-sri-lanka-duped-by-china-in-hambantota-port-101656205405799.html, 2022. [Accessed 03-05-2024].

Ravi Kanbur. The economics of international aid. *Handbook of the economics of giving, altruism and reciprocity*, 2:1559–1588, 2006.

David Halloran Lumsdaine. *Moral vision in international politics: the foreign aid regime, 1949-1989*. Princeton University Press, 1993.

OECD. International aid rises in 2023 with increased support to ukraine and humanitarian needs. https://www.oecd.org/newsroom/international-aid-rises-in-2023-with-increased-support-to-ukraine-and-humanitarian-needs.htm#:~:text=11%2F04%2F2024%20%2D%20International,data%20collected%20by%20the%20OECD, 2024. [Accessed 03-05-2024].

Carlos Seiglie. Altruism, foreign aid and humanitarian military intervention. *Conflict Management and Peace Science*, 17(2):207–223, 1999.

Olav Stokke. Aid and political conditionality: Core issues and state of the art. *Aid and political conditionality*, 82:1–87, 1995.

Victoria Williams. Foreign aid, Oct 2023. URL https://www.britannica.com/money/foreign-aid.